Grade 10 – Term 1: C9 Classroom Activity

Name: _____ Date: _____

Goal: Construct and interpret decision trees for staged filtering. Use supplied summaries (thousand USD). Maximax ranks Max Payoff highest first; Maximin ranks Min Payoff highest first; Minimax Regret ranks Max Regret lowest first; EMV ranks EMV highest first. Compare only survivors; eliminated alternatives never return. Keep all ties at each cutoff; report final ties. All statistics stay fixed, including regrets based on all six original alternatives.

A filter tree branches at each stage to survivors and eliminated alternatives; label each split with its rule. Use your notebook for full tables, trees, and extra working.

Guided Problems

1. City logistics policy.

Alternative	EMV	Max Payoff	Min Payoff	Max Regret
B_1	74.50	95.00	30.00	25.00
B_2	75.50	90.00	40.00	20.00
B_3	75.00	110.00	10.00	45.00
B_4	77.50	85.00	55.00	25.00
B_5	76.50	100.00	35.00	20.00
B_6	73.40	92.00	20.00	35.00

(a) Apply EMV: keep the best 4; then Maximin: keep the best 2; then Minimax Regret: select the best. Complete the table with values at every stage.

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$B_1, B_2, B_3, B_4, B_5, B_6$	—
1	EMV: keep best 4	$B_4(77.50)$, ...	_____
2	Maximin: keep best 2	_____	_____
3	Minimax Regret: select best	_____	_____

(b) Draw a tree from Step 0 using your table. At each stage, show a separate eliminated branch. Why cannot B_5 win the final regret comparison even though its Max Regret is 20?

2. Regional distributor expansion.

Alternative	EMV	Max Payoff	Min Payoff	Max Regret
X_1	80.50	130.00	20.00	40.00
X_2	85.25	120.00	40.00	20.00
X_3	77.75	140.00	10.00	50.00
X_4	91.25	110.00	60.00	30.00
X_5	84.75	125.00	45.00	20.00
X_6	84.15	118.00	35.00	25.00

(a) Maximax: keep the best 3. Complete the first split of your tree: survivors $X_3(140.00)$, _____, _____; eliminated _____.

(b) From those survivors, EMV: select the highest. Add the second split, showing all three EMVs, the two eliminated plans, and the final selection.

(c) X_4 has the highest original EMV. Why is it not the final choice?

Standard Problems

For each problem, construct a complete filtering table with the columns *# Step | Description | Left Alternatives | Eliminated Alternatives*. Include Step 0, numerical comparisons at every stage, and the final decision. Draw the matching filter tree, labeling the survivor and eliminated branches at every stage.

3. Trade finance guarantees.

Alternative	EMV	Max Payoff	Min Payoff	Max Regret
G_1	76.60	100.00	32.00	24.00
G_2	78.40	94.00	44.00	20.00
G_3	76.10	114.00	12.00	44.00
G_4	81.10	90.00	56.00	24.00
G_5	78.90	104.00	38.00	18.00
G_6	76.40	96.00	26.00	30.00

Apply these rules in order: (1) EMV: keep the best 5; (2) Maximax: keep the best 4; (3) Maximin: keep the best 2; (4) Minimax Regret: select the best. State when G_4 is eliminated and explain numerically why its original highest EMV cannot rescue it.

4. Insurance portfolio mixes.

Alternative	EMV	Max Payoff	Min Payoff	Max Regret
Y_1	86.40	136.00	28.00	36.00
Y_2	90.00	126.00	46.00	20.00
Y_3	82.10	146.00	14.00	50.00
Y_4	95.35	116.00	64.00	30.00
Y_5	88.35	130.00	48.00	19.00
Y_6	86.90	121.00	38.00	26.00

Apply these rules in order: (1) EMV: keep the best 3; (2) Minimax Regret: keep the best 2; (3) Maximin: select the best. State when Y_4 is eliminated. Explain why the final Maximin comparison cannot include its Min Payoff of 64.

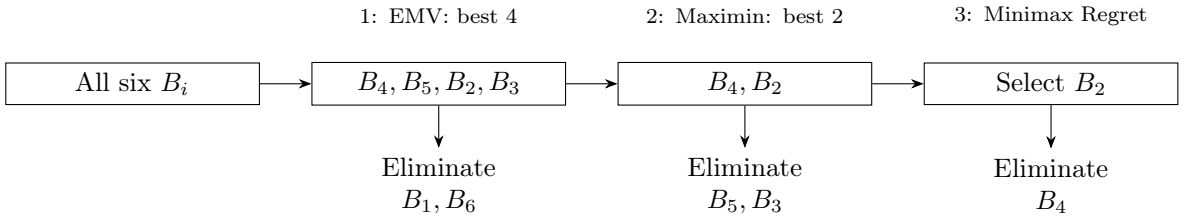
Solutions

Parentheses show the statistic used at that stage. Each elimination is new. All original summary values remain fixed. Downward tree branches show permanent eliminations; rightward branches show survivors. Tables give the numerical justification for each split.

1. City Logistics Policy

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{B_1, B_2, B_3, B_4, B_5, B_6\}$	—
1	EMV: keep the 4 highest EMV values.	$B_4(77.50)$, $B_5(76.50)$, $B_2(75.50)$, $B_3(75.00)$	$B_1(74.50)$, $B_6(73.40)$
2	Maximin: keep the 2 highest Min Payoff values.	$B_4(55.00)$, $B_2(40.00)$	$B_5(35.00)$, $B_3(10.00)$
3	Minimax Regret: select the lowest Max Regret value.	**Selected:** $B_2(20.00)$	$B_4(25.00)$

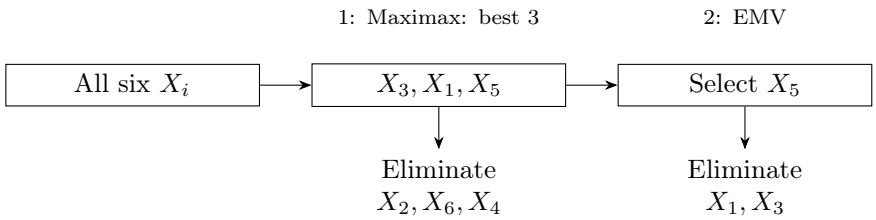
(a) EMVs rank $77.50 > 76.50 > 75.50 > 75.00 > 74.50 > 73.40$: cutoff 75.00. Surviving minima rank $55 > 40 > 35 > 10$: cutoff 40. Finally $20 < 25$, so select B_2. **(b)** B_5 was eliminated at Step 2 because $35 < 40$. Its regret 20 cannot restore it. Only B_4, B_2 enter Step 3. No cutoff or final ties occur.



2. Regional Distributor Expansion

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{X_1, X_2, X_3, X_4, X_5, X_6\}$	—
1	Maximax: keep the 3 highest Max Payoff values.	$X_3(140.00)$, $X_1(130.00)$, $X_5(125.00)$	$X_2(120.00)$, $X_6(118.00)$, $X_4(110.00)$
2	EMV: select the highest EMV value.	**Selected:** $X_5(84.75)$	$X_1(80.50)$, $X_3(77.75)$

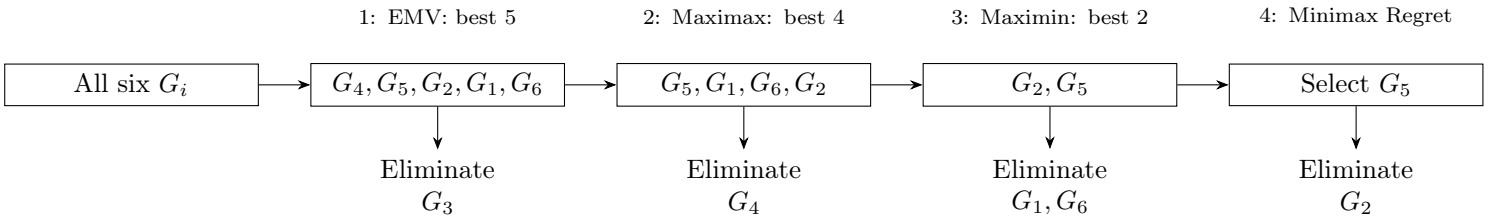
(a) Maxima rank $140 > 130 > 125 > 120 > 118 > 110$: cutoff 125. The blanks are $X_1(130.00)$ and $X_5(125.00)$; eliminate X_2, X_6, X_4. **(b)** Surviving EMVs rank $84.75 > 80.50 > 77.75$: select X_5; eliminate X_1, X_3. **(c)** X_4 has EMV 91.25, but its maximum $110 < 125$ eliminates it at Step 1. No cutoff or final ties occur.



3. Trade Finance Guarantees

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{G_1, G_2, G_3, G_4, G_5, G_6\}$	—
1	EMV: keep the 5 highest EMV values.	$G_4(81.10)$, $G_5(78.90)$, $G_2(78.40)$, $G_1(76.60)$, $G_6(76.40)$	$G_3(76.10)$
2	Maximax: keep the 4 highest Max Payoff values.	$G_5(104.00)$, $G_1(100.00)$, $G_6(96.00)$, $G_2(94.00)$	$G_4(90.00)$
3	Maximin: keep the 2 highest Min Payoff values.	$G_2(44.00)$, $G_5(38.00)$	$G_1(32.00)$, $G_6(26.00)$
4	Minimax Regret: select the lowest Max Regret value.	**Selected:** $G_5(18.00)$	$G_2(20.00)$

Step 1: $81.10 > 78.90 > 78.40 > 76.60 > 76.40 > 76.10$: eliminate G_3 below cutoff 76.40. Step 2: surviving maxima rank $104 > 100 > 96 > 94 > 90$: eliminate G_4 below 94. Step 3: surviving minima rank $44 > 38 > 32 > 26$: retain G_2, G_5. Step 4: $18 < 20$: select G_5. G_4 is eliminated at Step 2; its EMV 81.10 cannot override that Maximax filter. No cutoff or final ties occur.



4. Insurance Portfolio Mixes

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{Y_1, Y_2, Y_3, Y_4, Y_5, Y_6\}$	—
1	EMV: keep the 3 highest EMV values.	$Y_4(95.35)$, $Y_2(90.00)$, $Y_5(88.35)$	$Y_6(86.90)$, $Y_1(86.40)$, $Y_3(82.10)$
2	Minimax Regret: keep the 2 lowest Max Regret values.	$Y_5(19.00)$, $Y_2(20.00)$	$Y_4(30.00)$
3	Maximin: select the highest Min Payoff value.	**Selected:** $Y_5(48.00)$	$Y_2(46.00)$

Step 1: $95.35 > 90.00 > 88.35 > 86.90 > 86.40 > 82.10$: retain Y_4, Y_2, Y_5 at cutoff 88.35. Step 2: their regrets rank $19 < 20 < 30$: retain Y_5, Y_2 at cutoff 20 and eliminate Y_4. Step 3: $48 > 46$: select Y_5. Y_4 was eliminated at Step 2, so its minimum 64 is excluded from the final comparison. No cutoff or final ties occur.

